

Growing the Brownian CRT

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Based on joint work with Alessandra Caraceni, Nicolas Curien and William
Fleurat.

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Towards a CRT dynamic

- Question: Can we couple a sequence of “growing” $(T_n)_{n \geq 1}$ uniform binary trees of sizes $n \in \mathbb{N}$ and a process of Brownian CRT's $(\mathcal{T}_t)_{t \in [0,1]}$ of masses $t \in [0,1]$ such that

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- Remark: No topology specified.

Discrete trees

- Let's start with a definition of rooted planar binary trees.

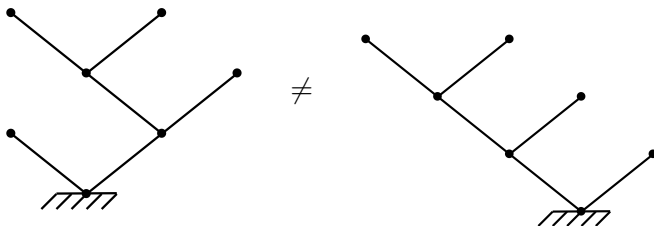


Figure: Two rooted planar binary trees of size 4 that are not equal.

Discrete trees

- We say that a sequence $(t_n)_{n \geq 1}$ is *growing* if for all $n \in \mathbb{N}$, we can obtain t_{n+1} from t_n by adding a cherry on a leaf of t_n .
- The reverse operation, i.e. cutting a cherry, is called *pruning*.

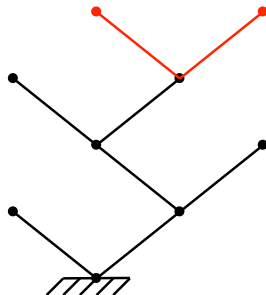


Figure: A cherry highlighted on a rooted planar binary tree.

Growing uniform binary trees

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- Caraceni, Curien, Stephenson ['24b] : Studied asymptotic properties of this algorithm.
- Caraceni, Curien, Fleurat, T. ['26] : Interpreted the previous 2 works as a leaf-growth algorithm on *labeled* binary trees.

Łuczak – Winkler and Caraceni – Stauffer pruning

- Start with a uniform binary tree T_{n+1} .

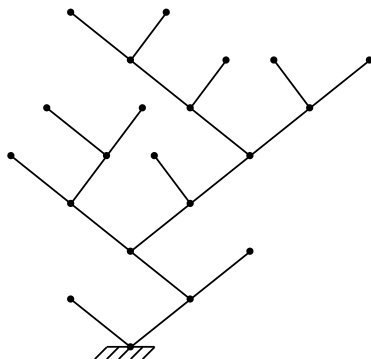


Figure: A uniform binary tree T_{11} .

Łuczak – Winkler and Caraceni – Stauffer pruning

- Start an exploration from the root, going left with probability $C(a, b)$ (and right with probability $1 - C(a, b)$), where a, b are the number of leaves in the left and right subtrees respectively.

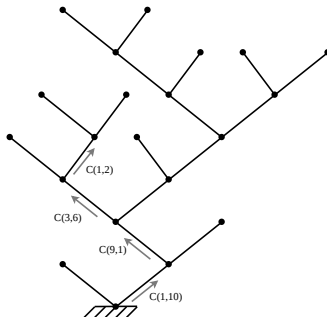


Figure: Such an exploration in the tree T_{11} .

Łuczak – Winkler and Caraceni – Stauffer pruning

- Once landed at the bottom of a cherry, prune.
- Given uniform T_{n+1} with $n + 1$ leaves, the pruned tree T_n is uniform with n leaves.

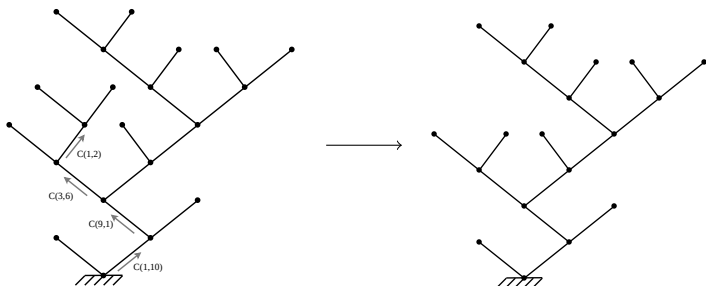


Figure: The tree T_{10} after pruning.

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Best-of-Three pruning algorithm

- Start with a uniform binary tree T_{n+1} .

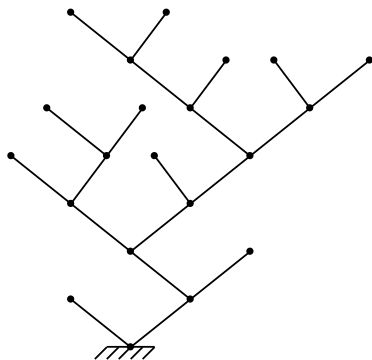


Figure: A uniform binary tree T_{11} .

Best-of-Three pruning algorithm

- Sample a uniform labeling ℓ of the leaves, yielding T_{n+1}^ℓ .

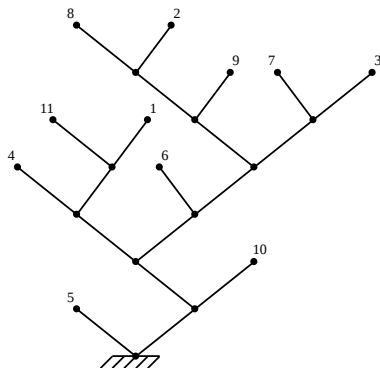


Figure: A uniform *labeled* binary tree T_{11}^ℓ .

Best-of-Three pruning algorithm

- Start at the root and follow the direction that has at least 2 smallest leaves until landing at the bottom of a cherry.

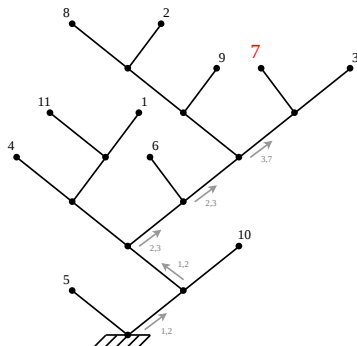


Figure: Following the majority of 3 smallest leaves.

Best-of-Three pruning algorithm

- Prune the cherry and keep the smallest label.

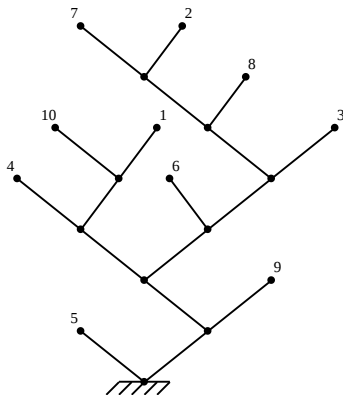


Figure: A binary tree T_{10}^ℓ .

Best-of-Three pruning algorithm

Lemma

Given uniform labeled T_{n+1}^ℓ , the tree T_n^ℓ obtained by the Best-of-Three pruning algorithm is uniform labeled with n leaves.

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- Forgetting the labels of T_n^ℓ , the tree T_n is uniform.
- The joint law of (T_n, T_{n+1}) is the same for both algorithms LW - CS and BoT !

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- “Independent” : Sample a *new*, independent labeling for T_n and perform the algorithm again to obtain T_{n-1} .

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- In the algorithm of Łuczak – Winkler and Caraceni – Stauffer, the labeling ℓ was implicitly resampled after every step.
- “Independent” : Sample a *new*, independent labeling for T_n and perform the algorithm again to obtain T_{n-1} .
- “Entropic” : Use the *same* labeling and apply the algorithm.

Using independent labeling

- Sample a new labeling ℓ' for T_n .

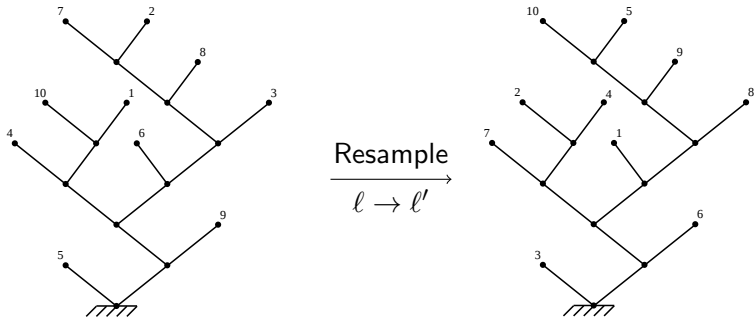


Figure: The binary tree T_{10}^ℓ given a *new* labeling, yielding $T_{10}^{\ell'}$.

Using independent labeling

- Perform the Best-of-Three pruning algorithm.

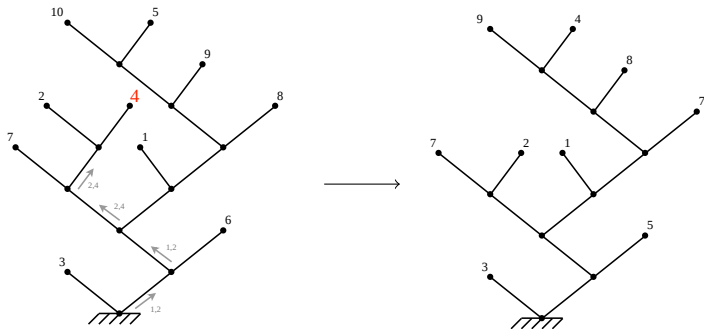


Figure: The Best-of-Three pruning algorithm applied to $T_{10}^{\ell'}$, yielding $\tilde{T}_9^{\ell'}$.

Using the same labeling

- Perform the Best-of-Three pruning algorithm on the tree T_n^ℓ where we kept the labeling.

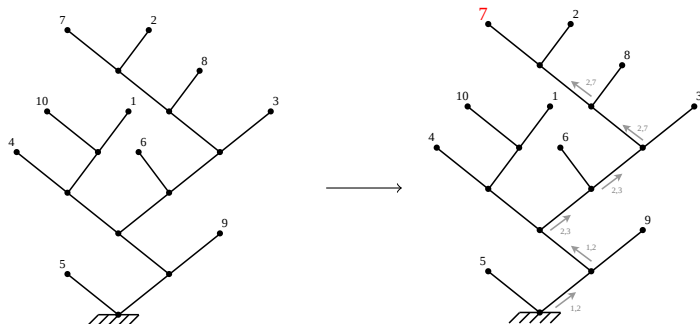


Figure: The leaf-deletion algorithm applied to T_{10}^ℓ , yielding T_9^ℓ .

Comparison

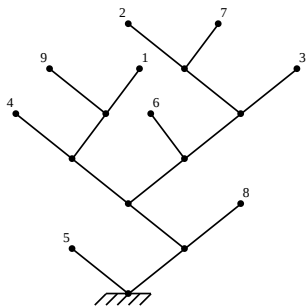


Figure: The tree T_9^ℓ obtained after applying twice the pruning algorithm using the same labeling.

Comparison

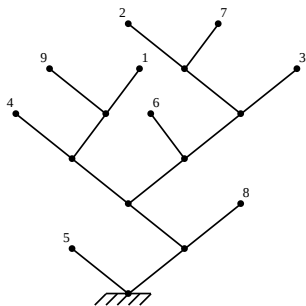


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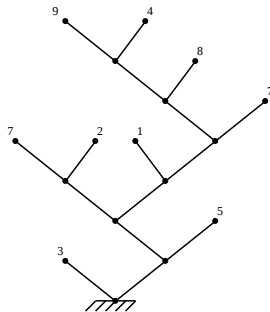
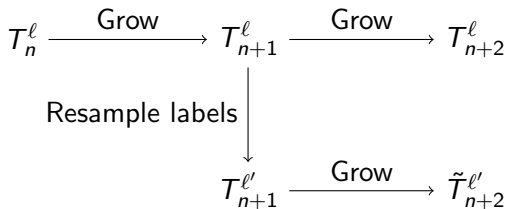


Figure: The tree $\tilde{T}_9^{\ell'}$ obtained after applying twice the pruning algorithm using independent labelings.

From pruning to leaf-growth

- By looking at independent or the same labeling at each step, we obtain different sequences for $(T_n)_{n \geq 1}$.



Result

Theorem (Caraceni, Curien, Fleurat, T. ('26+))

Let $(T_n)_{n \in \mathbb{N}}$ be the sequence of uniform binary trees constructed with the BoT leaf-growth algorithm using **independent** labelings. Then there exists a continuous process $(\mathcal{T}_t)_{t \in [0,1]}$ (constructed by Curien, Fleurat, T. in ['25]) such that

$$\left(\frac{1}{\sqrt{n}} T_{\lfloor tn \rfloor} : t \in [0, 1] \right) \xrightarrow[n \rightarrow \infty]{(d)} (\mathcal{T}_t)_{t \in [0,1]},$$

where $\mathcal{T}_t$ is a Brownian CRT of mass t .

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- The process $t \mapsto \mathcal{T}_t$ grows *globally* through a countable number of leaves simultaneously.
- Proof relies on the self-similar Markov tree construction of Bertoin, Curien, Riera ['24a].

Result

Theorem (Caraceni, Curien, Fleurat, T. ['26])

*Let $(T_n)_{n \in \mathbb{N}}$ be the sequence of uniform binary trees constructed with the BoT leaf-growth algorithm using **the same** labeling. Then there exists a continuous process $(\mathcal{T}_t)_{t \in [0,1]}$ such that*

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- The process $(\mathcal{T}_t)_{t \in [0,1]}$ grows *locally*, through a single point!
- The point through which $t \mapsto \mathcal{T}_t$ grows moves càdlàg over $\mathcal{T}$.

Simulations

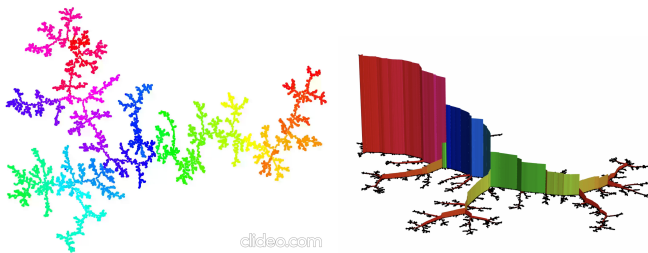


Figure: Second simulation inspired by figures in ['24a].

End

Thank you for your attention !